The empirical recurrence for OEIS A237310: recovering a conjecture that is not written down

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Abstract

OEIS A237310 records "Empirical recurrence of order 99", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 192$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 99, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 99 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A237310 is "Number of (n+1)X(2+1) 0..3 arrays with the maximum plus the lower median minus the minimum of every 2X2 subblock equal.". It has offset 1 and begins

$$1352, 12690, 120650, 1218584, 12256598, 126084160, 1291478032, 13373871324, \ \ldots$$

The entry states:

> Empirical recurrence of order 99 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 10:03:22, revision 7) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 192$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' =$ 192.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 384$ exact terms of the model returns order 99 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{99} c_i\, a(n-i),$$

c_1	47	c_{26}	−52165101548208819131517	c_{51}	−2359648350326545480144126
c_2	−593	c_{27}	−126350543310335108724885	c_{52}	−8320626538604320092104523
c_3	−6034	c_{28}	851068376436783331487254	c_{53}	61619811178057065628795089
c_4	209412	c_{29}	333526510219159125235454	c_{54}	−1867891183268481595568105
c_5	−881491	c_{30}	−9873585288230759105178400	c_{55}	−11814017507720354648296869
c_6	−23035500	c_{31}	9087130176979965896032100	c_{56}	104329233034582666803332959
c_7	256394283	c_{32}	83813612624501034566866652	c_{57}	158432077979878211341670629
c_8	781242020	c_{33}	−179791185493872220689232403	c_{58}	−26473419116166483070328798
c_9	−25503577126	c_{34}	−497610731655140475307309698	c_{59}	−11386079182261191523871474
c_{10}	60366302682	c_{35}	1945774718550558035416596813	c_{60}	454050434680874572314367956
c_{11}	1362328035883	c_{36}	1584542621753977898679434143	c_{61}	−6810535664075408385031563
c_{12}	−8507424430349	c_{37}	−14874364095813077429431507886	c_{62}	−55685783091988308872478876
c_{13}	−37791085970951	c_{38}	4478474970190647173940629019	c_{63}	345493571958827580736045785
c_{14}	495747420548899	c_{39}	84494740541842431761051848970	c_{64}	468309985030193286681724329
c_{15}	25514490336593	c_{40}	−102488830855968523069263178314	c_{65}	−57060165129395367462547960
c_{16}	−17745443909382442	c_{41}	−350771958449396261394042481624	c_{66}	−20767134309343048984299448
c_{17}	44916413110264706	c_{42}	803420184878356186621833577718	c_{67}	601623262946810460528197287
c_{18}	411819926950776362	c_{43}	939403711575755478792611852145	c_{68}	−7249191969570944623663573
c_{19}	−2136257744773231952	c_{44}	−4211526559171449178736106353547	c_{69}	−43563044188489971002018627
c_{20}	−5485056430624823245	c_{45}	−481983306032583590408609448959	c_{70}	219001513098650445168082962
c_{21}	58938498056748485864	c_{46}	16259191403089202148568653467170	c_{71}	204726550269304990910404735
c_{22}	−217854388333160284	c_{47}	−10302185820360006899712452807104	c_{72}	−20610412824618243012667945
c_{23}	−1115495243927207165666	c_{48}	−46498492951738205050163908862095	c_{73}	−4190048931508891167836495
c_{24}	1934651117399131451174	c_{49}	64397720112190798755012332350392	c_{74}	118928612719799592132107086
c_{25}	14804643982532595006130	c_{50}	91714187195956200367053336406828	c_{75}	−1909759564567967480898515

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 99, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $99+4$ terms removed returns order 99 as well, so $\deg q_{\min} = 99 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $99+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 99 that this sequence satisfies — and that family turns out to have exactly one member.

References

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